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Analytical estimates for deformation behavior of fluid-filled elastomers with random microstructures

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ABSTRACT

Macroscopic response of fluid filled elastomers with random microstructures, is obtained for aligned, cylindrical pores with circular cross-section. Results are derived for mono-disperse as well as poly-disperse pores, for Newtonian fluids. The obtained estimates are validated for plain strain compression loadings using numerical simulations performed in commercial multiphysics software COMSOL. Good agreement is observed between the analytical estimates and numerical solutions for different initial porosities and loading rates. Standard deviation in numerical results is seen to be high for polydisperse pores. Nonetheless, the mean response in numerical solutions matches well with analytical estimates. The obtained results can form the basis for design and fabrication of a new class of materials which have the potential to replace conventional spring–damper systems, particularly where space is limited.

1. Introduction

Porous elastomers play an important role in varied applications ranging from stretchable electronics [1], oil–water separation [2] and impact protection [3]. For low loading rate applications, a liquid inclusion is added to the porous elastomer so as to keep the composite soft and at the same also have improved strength. Liquid metal filled elastomers are used to create conductive polymers for applications in stretchable electronics [1,4,5]. Most literature on mechanical behavior of such fluid filled elastomers is based on experiments. The liquid is not allowed to flow out of the pores in such cases. In the present work, the behavior of fluid-filled elastomeric composites, where the fluid is allowed to flow in/out of the pore when the composites are loaded (see Fig. 1), is studied. Macroscopic behavior of such fluid-filled elastomers having random microstructure is estimated theoretically.

Estimates for effective behavior of composites having periodic microstructures has been obtained by Christensen [6]. Homogenization theories have been developed for composites having periodic microstructures in [7]. These theories have been applied to study the behavior of composites reinforced with periodically arranged fibers [8,9]. Random microstructures are easier to fabricate compared to periodic microstructures. Several studies have been done with regards to estimation of effective behavior of hyperelastic composites having random microstructures. Ponte Castañeda [10] proposed homogenization based estimates for effective behavior of composite having nonlinear elastic phases. Ponte Castañeda and Willis [11] studied the overall response of particulate composite having random distribution of inclusions. The Hashin–Shtrikhman [12,13] type estimates obtained for the composite were different from the Mori–Tanka [14] estimates because the latter does not account for spatial distribution of randomly oriented inclusions. Further, Ponte Castañeda's second order homogenization method [15–17] provides accurate estimates for the macroscopic response of composites and is therefore widely used. Using the SOH method, Lopez-Pamies [18] proposed estimates for effective behavior of porous hyperelastic materials having random microstructures. Idiart [19] used SOH to estimate the effective behavior of viscoplastic materials having isotropic distribution of spherical pores or rigid inclusions. Further estimates were obtained for the limiting case of ideal plastic material with low porosity. Moraleda et al. [20] studied the response of porous elastomers under finite deformation using numerical simulations. Porous elastomer consisted of random distribution of cylindrical voids. Monodisperse and polydisperse pores were considered. The results matched very well with the second order homogenization (SOH) based method with field fluctuations proposed by Lopez-Pamies and Castañeda [18]. Lopez-Pamies and Idiart [21] proposed exact result for estimating the macroscopic response of porous neo-Hookean solid using Hamilton–Jacobi equation. Tang et al. [22] proposed a renormalization method based two-scale homogenization approach for porous elastomers. The model parameters were obtained using numerical simulations of random microstructures. The theoretical estimates matched well with the numerical simulation results obtained for a Neo-Hookean elastomer. Guo et al. [23] obtained the response of

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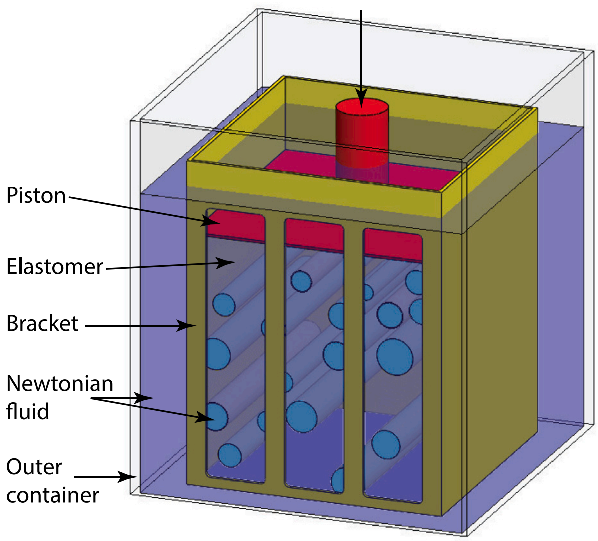

Fig. 1. Schematic representation of a fluid-filled elastomer subjected to plane strain compression. Polydispersed microstructure having random distribution of cylindrical pores is shown.

a hyperelastic elastomer reinforced with incompressible solid particles using numerical simulations. The response was obtained as a function of different initial volume fraction of particles and stiffness ratio of particles to matrix. The obtained data was fitted to a hyperelastic neo-Hookean model to model the composite as a homogeneous material. Jimenez [24] studied the effect of loading direction on isotropy of representative volume elements (RVEs) for fiber reinforced composite using numerical simulations. The parameters for study were RVE size and fiber volume fraction. It was observed that the homogenized response varied with the loading direction. Also the anisotropy under small stretch had very good correlation with anisotropy when large stretch was applied.

Based on the available literature, it is observed that there are no estimates for the macroscopic behavior and individual phase behavior of fluid-filled elastomeric composites with a random microstructure. Therefore, the present study fills the gap and provides estimates that can be used for the design of composites for the varied applications mentioned earlier. The effective behavior of fluid-filled elastomers having random but aligned pores and where the fluid is allowed to flow in and out of the pores, is studied here. The present work is an extension of previous work by the authors where a periodic microstructure was considered [25,26]. Random microstructure being more realistic, as is the case in bones, ceramics, woods and sponges, is considered here. In present work theoretical estimates are obtained for overall behavior of porous elastomers with randomly distributed monodisperse or polydisperse pores filled with fluid. Analytical estimates are validated using numerical simulations for multiple realizations having random but aligned pores. Numerical simulations have been carried out using COMSOL multiphysics software.

2. Numerical setup of fluid filled elastomer

2.1. Generation of microstructure

Each realization of random arrangement of pores consists of aligned pores randomly distributed in a cubical block. For monodispersed pores, the pore radius is set to 0.5 mm while for the polydispersed pores the pore radius was assumed to follow truncated Beta distribution with probability density function P and cumulative distribution function CDF defined as

$$\mathrm{P}(r;\alpha,\beta)=\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\frac{r^{(\alpha-1)}(1-r)^{(\beta-1)}}{(C(0.8;\alpha,\beta)-C(0.3;\alpha,\beta))}X(r) \tag{1}$$

$$\begin{aligned}\mathrm{CDF}(r;\alpha,\beta)&=\frac{\mathrm{B}(r;\alpha,\beta)}{(C(0.8;\alpha,\beta)-C(0.3;\alpha,\beta))\mathrm{B}(\alpha,\beta)}\\&=\frac{I_r(\alpha,\beta)}{C(0.8;\alpha,\beta)-C(0.3;\alpha,\beta)}\end{aligned} \tag{2}$$

where $\mathrm{B}(r;\alpha,\beta)$ is the incomplete beta function, $I_r(\alpha,\beta)$ is the regularized incomplete beta function and X(r) is defined as

$$X(r)=\begin{cases}1 & \text{if } 0.3\le r\le 0.8\\ 0 & \text{otherwise}\end{cases}$$

It is used to truncate the beta function. Fig. 2 shows (a) probability distribution and (b) cumulative probability distribution plot for the truncated beta distribution function taking $\alpha=\beta=5$. Initial pore radius r_o for the polydispersed pores ranges from 0.30 mm to 0.80 mm. All pores are taken to be cylindrical with circular cross-section initially. The size of elastomer block is 10 mm × 10 mm × 10 mm. The block size is taken to be an order of magnitude higher than the pore radii.

Fig. 3 shows two realizations for monodispersed and polydispersed random pores. The realizations shown are for initial porosity $f_o=0.19$.

2.2. Material behavior

2.2.1. Elastomer behavior

In the present work, strain energy density function for an incompressible neo-Hookean material is used to define the behavior of solid elastomer. The strain energy density function is defined as

$$W^{(s)}(\mathbf{F})=\frac{G}{2}(I_1-3), \tag{3}$$

where G is the shear modulus of homogeneous elastomer, $I_1=\mathrm{tr}(\mathbf{C})$ is the first principal invariant of right Cauchy–Green deformation tensor $\mathbf{C}=\mathbf{F}^T\mathbf{F}$, and $\mathbf{F}$ is the deformation gradient tensor. Displacements are primary variables to be solved for in the solid phase.

2.2.2. Fluid behavior

The pores are fully filled with an incompressible Newtonian fluid. Fluid flow in the pores is governed by incompressible Navier–Stokes equation. The continuity equation representing conservation of mass given by

$$\mathrm{div}\,\mathbf{v}=0 \tag{4}$$

and Navier–Stokes equation representing momentum conservation given by

$$\rho_f\frac{\mathrm{D}\mathbf{v}}{\mathrm{D}t}=-\nabla p+\mu\nabla^2\mathbf{v} \tag{5}$$

are solved in fluid domain. Here $\frac{\mathrm{D}\bullet}{\mathrm{D}t}$ represents total derivative of $\bullet$, p is pressure, $\mathbf{v}$ is velocity vector, ρ_f is fluid density and μ is fluid viscosity. In the fluid domain, the primary variables to be solved for are pressure and velocity. It is assumed that the solid and fluid do not chemically react with each other.

2.3. Boundary settings

2.3.1. Fluid domain

The fluid at fluid–solid interface is subjected to no-slip boundary condition. Pore ends are subjected to a constant pressure condition. The present case is an example of deformation induced flow. Deformation of the elastomer causes the fluid filled in pores to flow. The constant pressure at pore ends is applicable to fluid domain only whereas, no-slip condition at the interface is applicable to both solid and fluid domains.

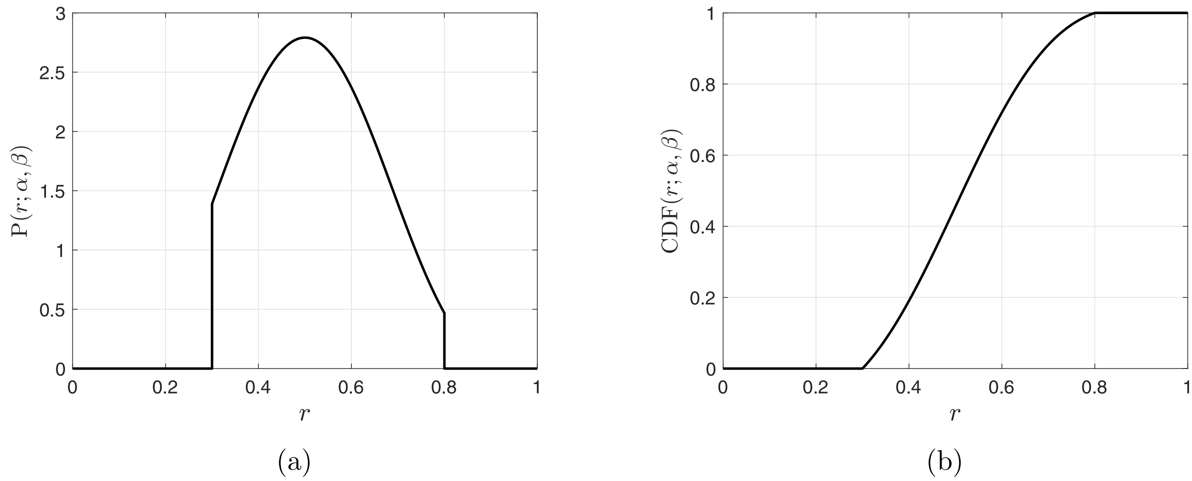

Fig. 2. (a) Probability distribution and (b) cumulative probability distribution for truncated beta distribution function for $\alpha = \beta = 5$.

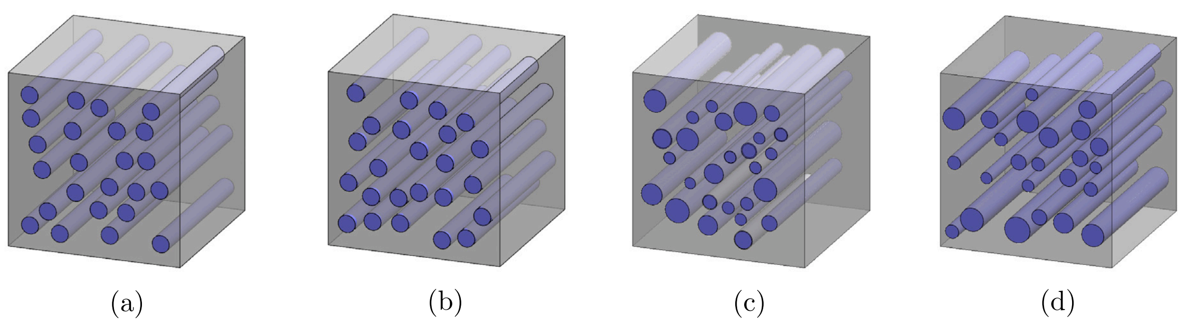

Fig. 3. Realizations of fluid filled elastomer composites having random but aligned (a, b) mono-dispersed and (c, d) poly-dispersed pores. The realizations shown are for initial porosity $f_o = 0.19$, initial pore radius $r_o = 0.5$ mm for mono-dispersed pores and cube side length of $a_o = 10$ mm. The parameters (α, β) are taken to be (5, 5) respectively in the present study. For poly-dispersed pores, the pore size ranges from 0.3 mm to 0.8 mm.

2.3.2. Solid domain

The unit cell boundary is subject to affine displacement boundary conditions. Displacements on the unit cell boundaries are given by

$$\mathbf{u} = (\overline{\mathbf{F}} - \mathbf{I})\mathbf{X} \tag{6}$$

where $\overline{\mathbf{F}}$, the applied macroscopic deformation gradient is given as

$$\overline{\mathbf{F}} = \begin{bmatrix} \bar{\lambda}_1 & 0 & 0 \\ 0 & \bar{\lambda}_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{7}$$

Although the elastomer is microscopically incompressible, the porous elastomer is compressible because of the presence of pores. The volume change of the porous elastomer during deformation is due to the change in volume of pores.

2.4. Theoretical estimates

Theoretical estimate for strain energy density of a porous elastomer having cylindrical pores was derived in a previous work by the authors [25] based on the composite cylindrical assemblage theory. The pore deformation obtained is used for estimating boundary conditions for fluid flow as the composite is deformed. The expression for strain energy density for the porous elastomer $\overline{W}_{\mathrm{SE}}$ is obtained as

$$\overline{W}_{\mathrm{SE}} = \frac{G}{2}\frac{(\bar{I}_1 - 1)(\bar{J} - 1)}{2\bar{J}}\log\left(\frac{f_o + \bar{J} - 1}{f_o\bar{J}}\right) + G(1 - f_o)\left(\frac{(\bar{I}_1 - 1)}{2\bar{J}} - 1\right). \tag{8}$$

where G is shear modulus of the elastomer, f_o is initial porosity in the elastomer and $\bar{I}_1$, $\bar{J}$ are the first and third invariant of macroscopic deformation gradient $\overline{\mathbf{F}}$. The average deformation gradient for the fluid phase $\overline{\mathbf{F}}^{(f)}$ is computed using the condition

$$\overline{\mathbf{F}} = (1 - f_o)\overline{\mathbf{F}}^{(s)} + f_o\overline{\mathbf{F}}^{(f)} \tag{9}$$

where $\overline{\mathbf{F}}^{(s)}$ is the average deformation gradient in solid phase. Using Eq. (9) and $\overline{\mathbf{F}}^{(s)}$ the components of $\overline{\mathbf{F}}^{(f)}$ can be estimated as

$$\overline{\mathrm{F}}^{(f)}_{11} = \bar{\lambda}^{(f)}_1 = \bar{\lambda}_1\sqrt{\frac{(f_o + \bar{J} - 1)}{f_o\bar{J}}}, \quad \overline{\mathrm{F}}^{(f)}_{22} = \bar{\lambda}^{(f)}_2 = \bar{\lambda}_2\sqrt{\frac{(f_o + \bar{J} - 1)}{f_o\bar{J}}} \tag{10}$$
$$\overline{\mathrm{F}}^{(f)}_{12} = \overline{\mathrm{F}}^{(f)}_{21} = \overline{\mathrm{F}}^{(f)}_{31} = \overline{\mathrm{F}}^{(f)}_{13} = \overline{\mathrm{F}}^{(f)}_{23} = \overline{\mathrm{F}}^{(f)}_{32} = 0, \quad \overline{\mathrm{F}}^{(f)}_{33} = \bar{\lambda}^{(f)}_3 = 1$$

A kinematically admissible velocity field in the fluid domain is proposed as [25]

$$v_1 = \frac{\dot{\bar{\lambda}}^{(f)}_1 x_1}{\bar{\lambda}^{(f)}_1}\left(A\left(1 - \frac{x_1^2}{\bar{\lambda}^{(f)2}_1 \mathrm{r}_o^2} - \frac{x_2^2}{\bar{\lambda}^{(f)2}_2 \mathrm{r}_o^2}\right) + 1\right) \tag{11a}$$

$$v_2 = \frac{\dot{\bar{\lambda}}^{(f)}_2 x_2}{\bar{\lambda}^{(f)}_2}\left(B\left(1 - \frac{x_1^2}{\bar{\lambda}^{(f)2}_1 \mathrm{r}_o^2} - \frac{x_2^2}{\bar{\lambda}^{(f)2}_2 \mathrm{r}_o^2}\right) + 1\right) \tag{11b}$$

$$v_3 = C\left(1 - \frac{x_1^2}{\bar{\lambda}^{(f)2}_1 \mathrm{r}_o^2} - \frac{x_2^2}{\bar{\lambda}^{(f)2}_2 \mathrm{r}_o^2}\right)\frac{x_3}{l_o}. \tag{11c}$$

Viscous dissipation density of the composite is estimated as

$$\dot{\overline{W}}_{\mathrm{VD}} = \frac{2\mu}{V_0}\int_{V_f}\left(\left(\frac{\partial v_1}{\partial x_1}\right)^2+\left(\frac{\partial v_2}{\partial x_2}\right)^2+\left(\frac{\partial v_3}{\partial x_3}\right)^2 + \frac{1}{2}\left[\left(\frac{\partial v_1}{\partial x_2}+\frac{\partial v_2}{\partial x_1}\right)^2+\left(\frac{\partial v_3}{\partial x_1}+\frac{\partial v_1}{\partial x_3}\right)^2+\left(\frac{\partial v_3}{\partial x_2}+\frac{\partial v_2}{\partial x_3}\right)^2\right]\right)dV \quad (12)$$

where μ is fluid viscosity, V_0 is the volume of composite and V_f is the volume of fluid domain. Using Eqs. (11a)–(11c) and (12) viscous dissipation density for monodispersed pores $\dot{\overline{W}}_{\mathrm{VD}}^{Mono}$, when $l_{\mathrm{o}}/r_{\mathrm{o}} \gg 1$ is obtained to be

$$\dot{\overline{W}}_{\mathrm{VD}}^{Mono} = \frac{\mu f_{\mathrm{o}}}{3}\left(\frac{2\,l_{\mathrm{o}}}{r_{\mathrm{o}}}\right)^2\left(\frac{\dot{\overline{\lambda}}_1^{(\mathrm{f})}}{\overline{\lambda}_1^{(\mathrm{f})}}+\frac{\dot{\overline{\lambda}}_2^{(\mathrm{f})}}{\overline{\lambda}_2^{(\mathrm{f})}}\right)^2\left(\frac{\overline{\lambda}_1^{(\mathrm{f})}}{\overline{\lambda}_2^{(\mathrm{f})}}+\frac{\overline{\lambda}_2^{(\mathrm{f})}}{\overline{\lambda}_1^{(\mathrm{f})}}\right) \quad (13)$$

where l_{o} and r_{o} are pore half-length and initial pore radius respectively and $\dot{\overline{\lambda}}_i^{(\mathrm{f})}$, $\overline{\lambda}_i^{(\mathrm{f})}$ are the rate of pore deformation and pore deformation along the ith principal direction (i = 1, 2). The above expression (Eq. (13)) is applicable in case of monodispersed pores. For polydispersed pores the macroscopic viscous dissipation density is estimated as

$$\dot{\overline{W}}_{\mathrm{VD}}^{Poly} = \sum_{i=1}^{P}\frac{\mu f_{\mathrm{o}}}{3}\left(\frac{2\,l_{\mathrm{o}}}{r_{\mathrm{o}\,i}}\right)^2\left(\frac{\dot{\overline{\lambda}}_1^{(\mathrm{f})}}{\overline{\lambda}_1^{(\mathrm{f})}}+\frac{\dot{\overline{\lambda}}_2^{(\mathrm{f})}}{\overline{\lambda}_2^{(\mathrm{f})}}\right)^2\left(\frac{\overline{\lambda}_1^{(\mathrm{f})}}{\overline{\lambda}_2^{(\mathrm{f})}}+\frac{\overline{\lambda}_2^{(\mathrm{f})}}{\overline{\lambda}_1^{(\mathrm{f})}}\right)\times\frac{V(r_{\mathrm{o}\,i})}{\pi\, r_{\mathrm{o}\,i}^2\, 2l_{\mathrm{o}}} \quad (14)$$

where $V(r_{\mathrm{o}\,i})$ is the volume of pores of radius $r_{\mathrm{o}\,i}$ obtained statistically, when the pore size distribution follows Beta distribution and P is the number of divisions of the range of pore radii considered. Step size for each division is calculated as

$$\Delta r = \frac{r_{o\,max}-r_{o\,min}}{P} \quad (15)$$

in the present case $r_{o\,max}$, $r_{o\,min}$ are 0.8 mm and 0.3 mm respectively and P = 100. Mean radius for the ith step $r_{\mathrm{o}\,i}$, is calculated as

$$r_{oi} = r_{o\,min} + (i - 1/2)\Delta r \quad (16)$$

2.5. Numerical simulation setup

Fully coupled fluid–structure interaction simulations are carried out in COMSOL multiphysics. Navier–Stokes equation is considered for fluid flow. Fluid domain is considered as a deforming domain because its boundary keeps moving during the simulation. Simulations were carried out for different mesh sizes and appropriate mesh is chosen to reduce the computation time.

Quadratic serendipity elements are used to discretize the solid domain. In the fluid domain, velocity field is discretized using second-order elements whereas pressure field is discretized using linear elements. Backward difference method is used for time stepping. PARDISO, a direct solver available within COMSOL multiphysics is used. In the numerical simulations, primary variables for the fluid domain are solved using Eulerian formulation, while for the solid domain Lagrangian formulation is used. For proper transfer of motion from solid to fluid, arbitrary Lagrangian–Eulerian method is used at the fluid–structure interface. Velocity continuity is enforced at fluid–solid interface.

3. Results and discussion

In this section, we compare the results from analytical framework and results from numerical simulations of composite having an elastomer block with random but aligned arrangement of pores filled with Newtonian fluid.

3.1. Random microstructure with monodisperse pores

3.1.1. Effect of porosity

Strain energy density of the elastomeric composite is plotted in Fig. 4(a). Analytical estimates are in excellent agreement with the numerical simulation results obtained for a porous elastomer having random microstructure with monodispersed pores. Fig. 4(b) shows a comparison of the composite's viscous dissipation density obtained from numerical simulation with that from analytical estimate. The standard deviation for viscous dissipation density is very small which shows that random position of monodisperse pores has negligible effect on it. Fig. 4(c) shows the variation of stress with applied strain for monodispersed microstructure and analytical results. The numerical results have an excellent agreement with analytical results. The theoretical estimates for overall response are determined using rate of change of strain energy density and viscous dissipation density. Since, the analytical results for strain energy density and viscous dissipation density are in excellent agreement with the numerical results it is expected to give a good estimate for overall response of the composite. For a given deformation, response of the composite decreases with increase in porosity. An increase in porosity means reduction in elastomer volume, which is the main load bearing element in case of low loading rates as is the case in present section and hence the average stress decreases with increase in porosity.

3.1.2. Effect of loading rate

Fig. 5 shows the comparison of strain energy density, viscous dissipation density and overall response of the composite as estimated using theoretical estimates and from numerical simulations for different loading rates at a constant initial porosity f_{o} = 0.16. Fig. 5(a) shows a comparison of numerical results and theoretical estimates for stored energy density in the composite under different loading rates $\dot{\overline{\lambda}} = -0.05$ s^{-1}, -0.50 s^{-1} and -5.00 s^{-1}. The results for different values of $\dot{\overline{\lambda}}$ overlap showing that the stored energy density remains unaffected by the loading rate. It is also seen from Eq. (8) that the homogenized stored energy density function for the porous elastomer is independent of the loading rate and only depends on applied deformation. Fig. 5(b) shows a comparison of viscous dissipation density in the composite under the loading rates $\dot{\overline{\lambda}} = -0.05$ s^{-1}, -0.50 s^{-1}, -5.00 s^{-1}. As can be seen from the figure, the dissipation density varies as square of the loading rate for Newtonian fluids. Fig. 5(c) shows the comparison for overall response of the composite. The response for $\dot{\overline{\lambda}} = -0.05$ s^{-1} and -0.50 s^{-1} overlap because at low loading rate the contribution of viscous dissipation in fluid to overall response is small and hence negligible in comparison to the contribution of rate of change of strain energy density of elastomer. As the loading rate increases, the rate of change of strain energy density of elastomer remains constant but the dissipation in fluid increases and it becomes considerable in comparison to the rate of change of strain energy density as is evident from the plot for $\dot{\overline{\lambda}} = -5.00$ s^{-1}.

3.2. Random microstructure with polydisperse pores

3.2.1. Effect of porosity

Fig. 6 shows the comparison of numerical results and theoretical estimate for strain energy density, viscous dissipation density and overall response of the composite having a random but aligned distribution of poly-dispersed pores. The results are plotted for loading rate $\dot{\overline{\lambda}} = -0.05$ s^{-1} and initial porosity f_{o} = 0.13, 0.16 and 0.19. Fig. 6(a) shows the comparison for strain energy density of the composite. It is observed that the theoretical estimates are slightly higher than the results obtained from numerical simulations. Also the standard deviation is very small which shows the effect of pore size distribution is very small on the strain energy density of the composite. Fig. 6(b) shows the comparison for viscous dissipation density in the composite as a function of applied stretch for different initial porosity values. Error bar shows the standard deviation for numerical simulation results. The dissipation density decreases with increase in porosity but increases with applied deformation. The theoretical estimates have good match with numerical simulation results but they slightly under-predict the dissipation density with deformation. Fig. 6(c) shows the comparison for overall response of the composite having poly-dispersed pores. The

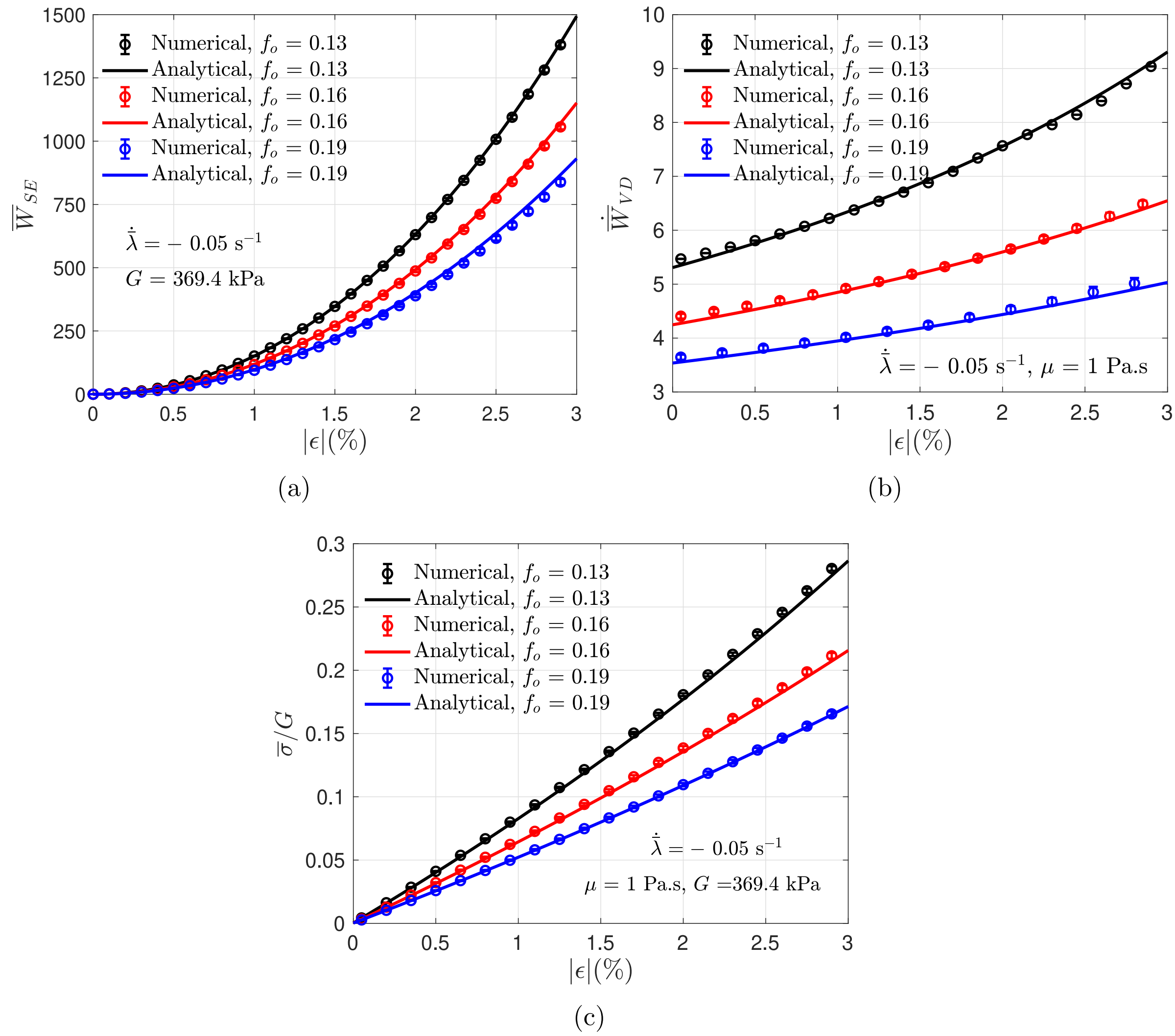

Fig. 4. Comparison of theoretical estimates for monodispersed fluid-filled composite with that from numerical simulation for (a) macroscopic stored energy density, (b) macroscopic viscous dissipation density and (c) effective behavior of the composite. The results have been plotted for initial porosity f_o = 0.13, 0.16 and 0.19, loading rate $\dot{\bar{\lambda}} = -0.05$ s^{-1}.

overall response is dominated by the elastomer with a small fraction from the dissipation effect. The theoretical estimates have a good match with the numerical simulation results.

3.2.2. *Effect of loading rate*

Fig. 7 shows the effect of applied loading rate on the overall response of the composite and on the strain energy density and viscous dissipation of the composite. Here, we compare the strain energy density, viscous dissipation density and overall response of the composite having polydispersed pores. The loading rates considered are −0.05 s^{-1}, −0.50 s^{-1} and −5.00 s^{-1} and the initial porosity in the elastomer f_o = 0.16. Polydispersity of pores provides an insight regarding effect of pore size scatter on the composite behavior. Fig. 7(a) shows the comparison for strain energy density of the composite. As already discussed, the strain energy density is independent of loading rate and only depends on applied deformation and initial porosity, the curves overlap for different loading rates considered. Also the pore size distribution has no effect on strain energy density of the composite. The theoretical estimates match well with numerical simulations. Fig. 7(b) shows the comparison of viscous dissipation density for the composite under different loading rates. As can be seen the dissipation density increases by 2 orders when the loading rate increases by a factor of 10. The numerical simulation results compare well with the theoretical estimates. Fig. 7(c) shows a comparison of the overall response of composite consisting of polydispersed pores as a function of applied deformation. The response of the composites for loading rates −0.05 s^{-1} and −0.50 s^{-1} are close to each other since they are dominated by the elastomer response. The deviation for theoretical results is smaller as compared to numerical simulation results. The response in the case of loading rate −5.00 s^{-1} is considerably higher than that for −0.05 s^{-1}, −0.50 s^{-1}. Therefore, it can be inferred that for loading rate higher than −1.00 s^{-1}, the dissipation effect starts to have an appreciable role in overall response of the composite for phase properties considered. The theoretical estimates and numerical simulation results have a very good match in the case of polydispersed pores too.

3.3. *Local response of the elastomer*

In this section we compare the von-Mises stress obtained from the numerical simulations carried out for both monodispersed and

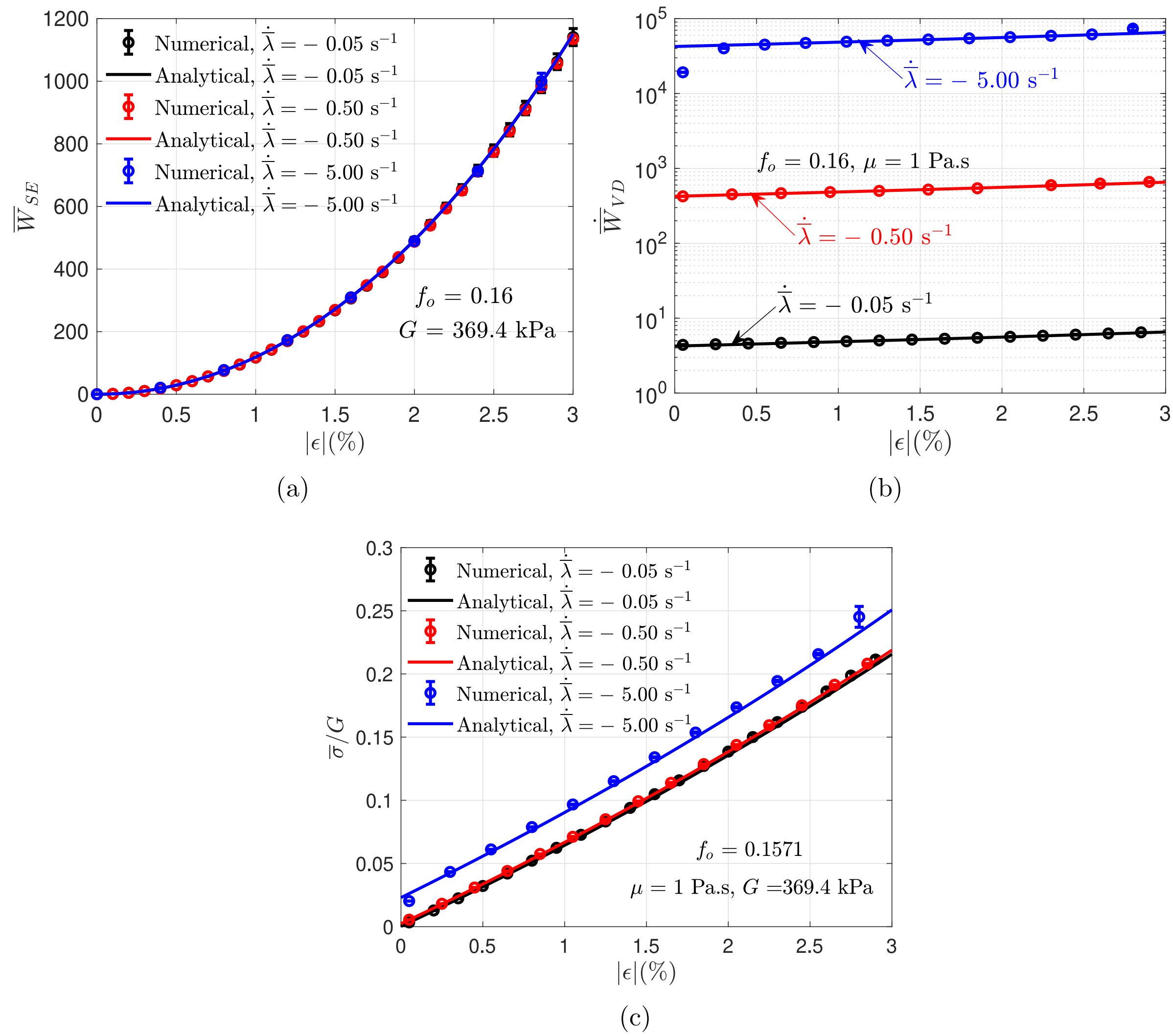

Fig. 5. Comparison of theoretical estimates for monodispersed fluid-filled composite with that from numerical simulation for (a) macroscopic stored energy density, (b) macroscopic viscous dissipation density and (c) effective behavior of the composite. The results have been plotted for initial porosity $f_o = 0.16$, loading rate $\dot{\bar{\lambda}} = -0.05$ s^{-1}, -0.50 s^{-1}, -5.00 s^{-1}.

polydispersed case. Fig. 8 shows contour plot of von-Mises stress in the cross-section plane located at $z = l_o/2$. In the mono-dispersed pores the pore radius is set at 0.5 mm and in the case of poly-dispersed pores it varies from 0.30 mm to 0.80 mm. In both the cases initial porosity $f_o = 0.16$, macroscopic loading rate $\dot{\bar{\lambda}} = -0.05$ s^{-1}, -5.00 s^{-1} and applied strain $|\epsilon| = 2\%$. It is observed from Figs. 8(a) and 8(b) that the peak stress in the case of mono-dispersed pores is approximately 1.5×10^5 Pa whereas that in the case of poly-dispersed pores (Figs. 8(c) and 8(d)) is approximately 1.9×10^5 Pa. Also it can be seen that the peak stress is unaffected by the applied loading rate. Peak stresses are observed in the region between the pores close to the pore boundary. This is due to the fact that during uniaxial plane strain compression the pore starts becoming elliptical and hence stress near the pore boundary is higher than any other region in the elastomer.

Conclusions

The present study deals with the comparison of numerical simulation results for fluid-filled elastomeric composite having random microstructure with the analytical estimates derived for a random porous composite using the concept of composite cylindrical assemblage. The composite consists of an elastomer having random but aligned pores filled with a Newtonian fluid. The study considers random microstructure with monodisperse and polydisperse pores. The random pore sizes are taken to follow truncated Beta distribution. The effect of initial porosity and loading rate have been investigated in the present study. The results obtained using theoretical estimates match very well with the numerical simulation results. The results for strain energy density are in excellent agreement with the numerical results. The results for viscous dissipation density show that it decreases with increase in porosity but increases with increase in applied deformation. The overall response results show that for low loading rates ($\lessapprox 1$ s^{-1}) the overall response of the composite is dictated by the elastomer behavior and for loading rates above 1 s^{-1} the viscous dissipation effect also plays its part in the overall response. The stress distribution plots show that pore location and undeformed pore size have a significant effect on the stress generated in the elastomer. Also the peak stresses remain unaffected by the applied loading rate. The present study validates the derived estimates which can be used for estimating the macroscopic response of fluid-filled elastomeric composite having

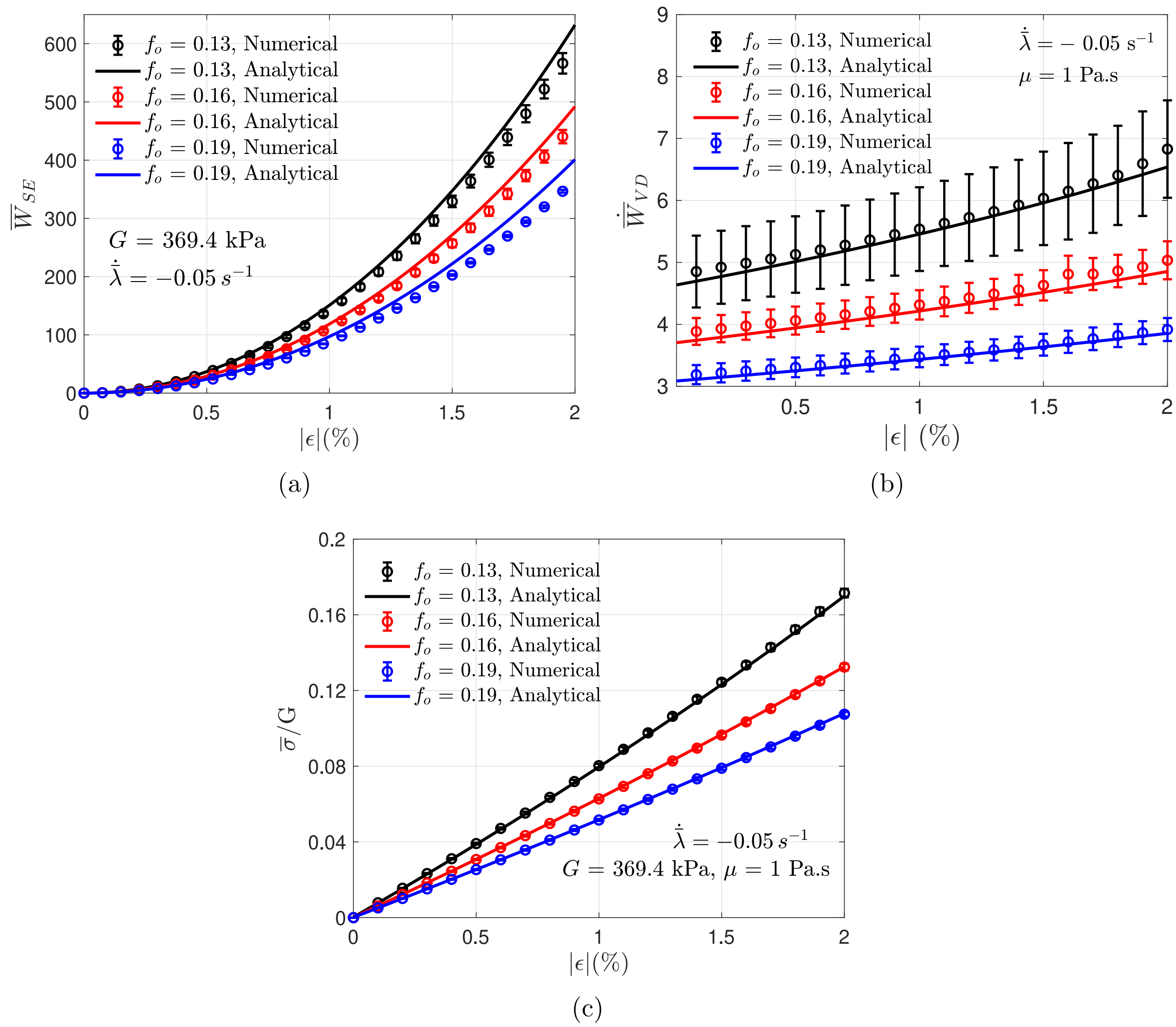

Fig. 6. Comparison of theoretical estimates for polydispersed fluid-filled composites with that from numerical simulation for (a) macroscopic stored energy density, (b) macroscopic viscous dissipation density and (c) effective behavior of the composite. The results have been plotted for initial porosity f_o = 0.13, 0.16, 0.19 and loading rate $\dot{\bar{\lambda}} = -0.05$ s^{-1}.

random microstructure. The equivalent spring stiffness and damping coefficient of the fluid-filled elastomeric composite can be estimated by modeling the composite as a Kelvin–Voigt element. The expressions obtained for equivalent spring stiffness and equivalent damping stiffness will be a function of loading, phase behavior, and the microstructure. Therefore, a few parameters can be tuned while others are fixed to obtain the desired stiffness and damping.

CRediT authorship contribution statement

Vivek Singh: Conceptualization, Methodology, Software, Validation, Formal analysis, Visualization, Investigation, Writing – original draft, Writing – review & editing. **Vikranth Racherla:** Conceptualization, Methodology, Visualization, Investigation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix A. Theoretical formulation

A kinematically admissible deformation field, is assumed such that

$$x_i = \psi_i(R)\, X_i, \quad R = \sqrt{X_1^2 + X_2^2} \quad (i = 1, 2) \quad \text{and there is no sum on } i \tag{17}$$

where $\mathbf{x}$ denotes the position of a material point in the solid phase in deformed configuration, $\mathbf{X}$ denotes the position of the corresponding material point in initial/undeformed configuration, and R is the radial distance of the point from pore center in the undeformed configuration. Here, $x_3 = X_3$. The components X_i $(i = 1, 2)$ in cylindrical coordinate system are given as

$$X_1 = R\cos\theta, \qquad X_2 = R\sin\theta \tag{18}$$

where $0 \le \theta \le 2\pi$. Components of the local deformation gradient in the elastomeric matrix can be obtained as

$$\mathrm{F}_{ij}^{(s)} = \psi_i' \frac{X_i X_j}{R} + \psi_i\, \delta_{ij}, \qquad i, j = 1, 2 \quad (\text{no sum on } i) \tag{19}$$

where the $()'$ symbol denotes derivative with respect to R. The elastomer is taken to be locally incompressible, therefore,

$$J^{(s)} \equiv \det\mathbf{F}^{(s)} = 1 \tag{20}$$

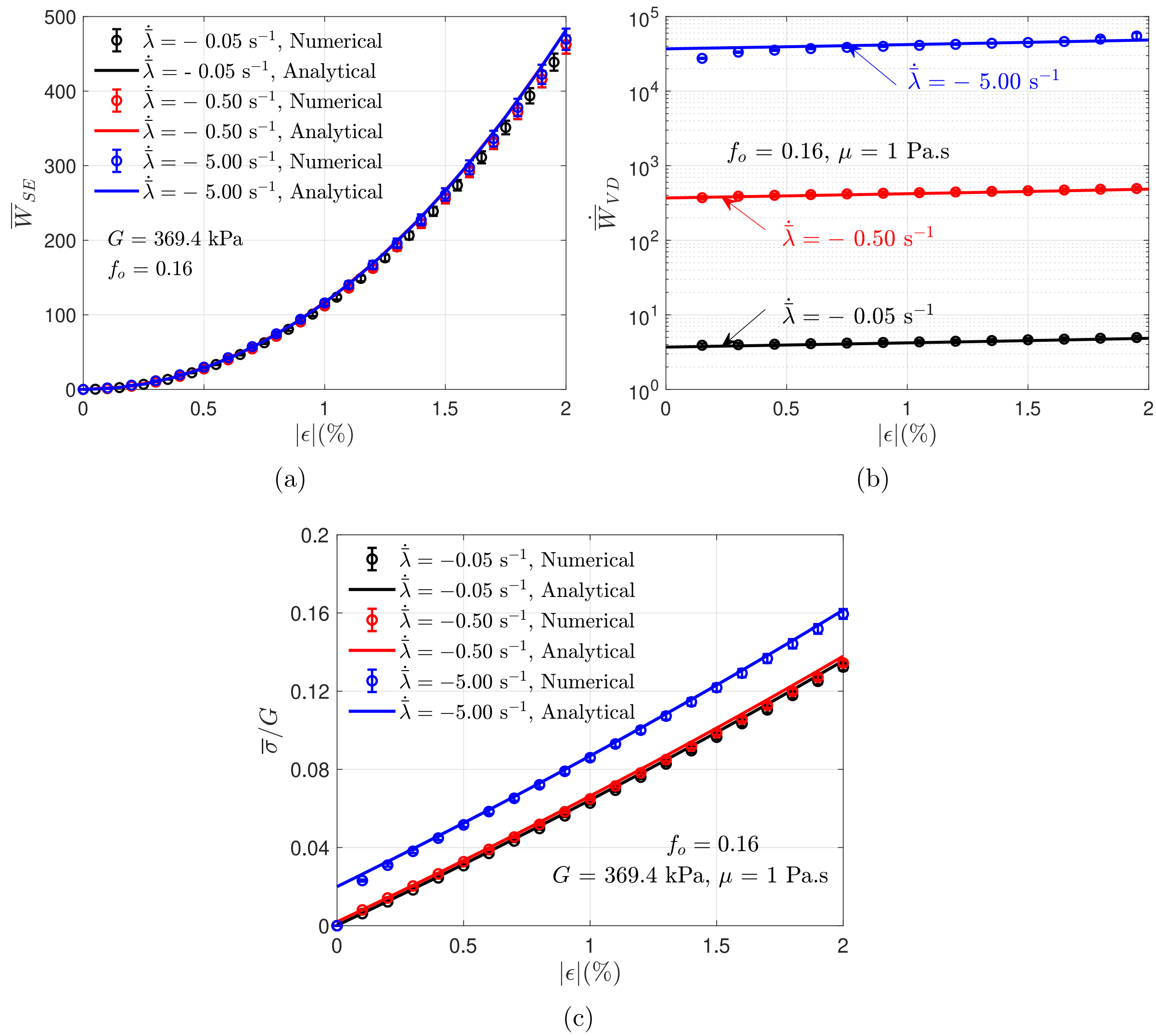

Fig. 7. Comparison of theoretical estimates for polydispersed fluid-filled composites with that from numerical simulation for (a) macroscopic stored energy density, (b) macroscopic viscous dissipation density and (c) effective behavior of the composite. The results have been plotted for initial porosity $f_o = 0.16$, loading rate $\dot{\lambda} = -0.05\ \mathrm{s}^{-1}, -0.50\ \mathrm{s}^{-1}$ and $-5.00\ \mathrm{s}^{-1}$.

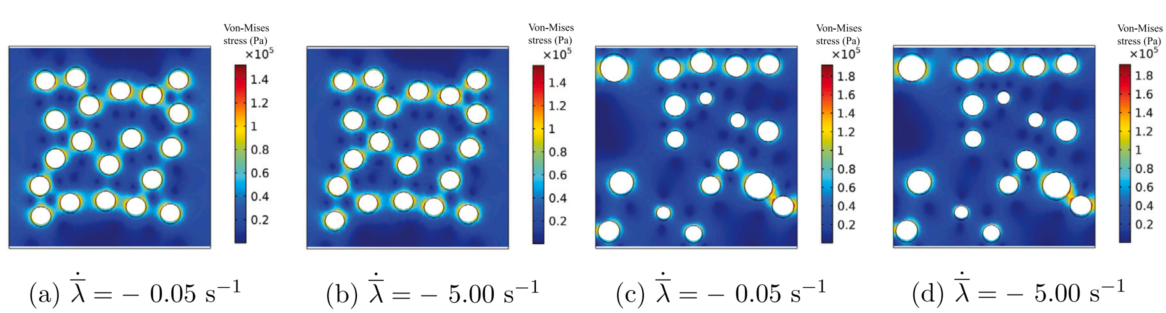

Fig. 8. Contour plots for Von-Mises stress in the cross-section plane located at $z = l_o/2$. Figures show randomly arranged (a, b) mono-dispersed pores and (c, d) poly-dispersed pores. The plots are obtained at $|\epsilon| = 2\%$, $\dot{\lambda} = -0.05\ \mathrm{s}^{-1}$ (a, c) and $-5.00\ \mathrm{s}^{-1}$ (b, d), $f_o = 0.16$.

From Eq. (20), it can be shown that at all points in the solid phase

$$\psi_1\psi_2 + R\,\psi_1\psi_2' = 1 \quad \text{and} \qquad \psi_1'\psi_2 - \psi_1\psi_2' = 0 \tag{21}$$

The two unknown functions ψ_1 and ψ_2 can be obtained by solving the system of differential equations in Eq. (21). To solve for these, we use the boundary conditions. The circular outer surface (with radius R_o) in initial configuration transforms into an ellipse with semi axes $\overline{\lambda}_1 R_o$, $\overline{\lambda}_2 R_o$ under the applied macroscopic deformation $\overline{\mathbf{F}}$. Using Eq. (17), we get

$$\psi_i|_{R=R_o} = \overline{\lambda}_i \qquad (i = 1,2) \tag{22}$$

Using Eq. (22) and system of equations in Eq. (21), it can be shown that

$$x_i = \frac{\overline{\lambda}_i}{\overline{J}^{1/2}}\left[1 + \left(\frac{R_o}{R}\right)^2(\overline{J} - 1)\right]^{1/2} X_i, \qquad (i = 1,2) \quad (\text{no sum on } i) \tag{23}$$

where $\overline{J} = \overline{\lambda}_1\overline{\lambda}_2$ and $x_3 = X_3$. Using Eqs. (19) and (23), the components of the local deformation gradient tensor $\mathbf{F}^{(s)}$ in the elastomeric matrix, can be expressed as

$$\begin{aligned} \mathrm{F}^{(s)}_{ij} &= \frac{\overline{\lambda}_i}{\overline{J}^{1/2}}\left(\frac{1-\psi^2}{\psi R^2}X_iX_j + \psi\delta_{ij}\right) \quad (i,j = 1,2) \qquad (\text{no sum on } i) \\ &\mathrm{F}^{(s)}_{31} = \mathrm{F}^{(s)}_{13} = \mathrm{F}^{(s)}_{23} = \mathrm{F}^{(s)}_{32} = 0, \quad \mathrm{F}^{(s)}_{33} = 1 \end{aligned} \tag{24}$$

where $\psi = \left[1 + \left(R_o/R\right)^2(\overline{J} - 1)\right]^{1/2}$. The average deformation gradient in the solid phase is estimated as

$$\overline{\mathbf{F}}^{(s)} = \frac{1}{\pi(R_o^2 - R_i^2)}\int_{R_i}^{R_o} R\,dR\int_0^{2\pi}\mathbf{F}^{(s)}\,d\theta \tag{25}$$

Using Eqs. (9) and (25) the components of $\overline{\mathbf{F}}^{(f)}$ are given in Eq. (10). The derivatives $\dot{\overline{\lambda}}{}^{(f)}_1$ and $\dot{\overline{\lambda}}{}^{(f)}_2$ are obtained as

$$\begin{aligned} \dot{\overline{\lambda}}{}^{(f)}_1 &= \frac{2\dot{\overline{\lambda}}_1\overline{J}(f_o + \overline{J} - 1) + \overline{\lambda}_1(1 - f_o)\dot{\overline{J}}}{2\,f_o^{1/2}\,\overline{J}^{3/2}\sqrt{f_o + \overline{J} - 1}}, \\ \dot{\overline{\lambda}}{}^{(f)}_2 &= \frac{2\dot{\overline{\lambda}}_2\overline{J}(f_o + \overline{J} - 1) + \overline{\lambda}_2(1 - f_o)\dot{\overline{J}}}{2\,f_o^{1/2}\,\overline{J}^{3/2}\sqrt{f_o + \overline{J} - 1}} \end{aligned} \tag{26}$$

Appendix B. Viscous dissipation in the fluid

Cauchy stress components $\boldsymbol{\sigma}_{ij}$ at any point in the fluid domain are given by

$$\boldsymbol{\sigma}_{ij} = -p\,\delta_{ij} + 2\mu\mathbf{D}_{ij} \quad (i,j = 1,2,3) \tag{27}$$

where p is the hydrostatic pressure, μ is the fluid viscosity and δ is kronecker delta. Rate of stretch tensor $\mathbf{D}$ is given by

$$\mathbf{D} = \frac{1}{2}\left[\frac{\partial\mathbf{v}}{\partial\mathbf{x}} + \left(\frac{\partial\mathbf{v}}{\partial\mathbf{x}}\right)^T\right] \tag{28}$$

Thus, the local dissipation density in the fluid is given by

$$\dot{W}^{(f)} = \boldsymbol{\sigma} : \mathbf{D} = 2\,\mu\,\mathbf{D}_{ij}\,\mathbf{D}_{ij} \tag{29}$$

where

$$\begin{aligned} \mathbf{D}_{ij}\mathbf{D}_{ij} = {}&\left(\frac{\partial v_1}{\partial x_1}\right)^2 + \left(\frac{\partial v_2}{\partial x_2}\right)^2 + \left(\frac{\partial v_3}{\partial x_3}\right)^2 + \frac{1}{2}\left[\left(\frac{\partial v_1}{\partial x_2} + \frac{\partial v_2}{\partial x_1}\right)^2\right. \\ &\left. + \left(\frac{\partial v_1}{\partial x_3} + \frac{\partial v_3}{\partial x_1}\right)^2 + \left(\frac{\partial v_2}{\partial x_3} + \frac{\partial v_3}{\partial x_2}\right)^2\right] \end{aligned}$$

Since,

$$\frac{\partial v_1}{\partial x_3} = \frac{\partial v_2}{\partial x_3} = 0$$

the expression for local viscous dissipation density becomes

$$\begin{aligned} \dot{W}^{(f)} = 2\mu\Bigg(&\left(\frac{\partial v_1}{\partial x_1}\right)^2 + \left(\frac{\partial v_2}{\partial x_2}\right)^2 + \left(\frac{\partial v_3}{\partial x_3}\right)^2 \\ &+ \frac{1}{2}\left[\left(\frac{\partial v_1}{\partial x_2} + \frac{\partial v_2}{\partial x_1}\right)^2 + \left(\frac{\partial v_3}{\partial x_1}\right)^2 + \left(\frac{\partial v_3}{\partial x_2}\right)^2\right]\Bigg) \end{aligned} \tag{30}$$

Using Eq. (30), the viscous dissipation density can be written as

$$\dot{\overline{W}}_{VD} = \frac{1}{V_o}\int_{V_f}\dot{W}^{(f)}dV \tag{31}$$

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